

W-ACT: Predictive Grounding in Action Chunking Transformers via Latent-Space World Models

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Abstract—The industrial sector faces a significant skills gap, compounded by the limitations of conventional training and the inherent brittleness of imitation learning architectures. While AR and digital twins offer potential for immersive instruction, current visuomotor policies often lack the predictive grounding required to respect real-world physical constraints. This paper proposes W-ACT (World-Model Augmented Action Chunking Transformer), a framework integrating a high-precision controller with a latent-space world model. By leveraging the V-JEPA-2 architecture and the EgoDex dataset, the proposed approach regularises policy output through predictive consistency, ensuring action sequences are both spatially accurate and physically feasible. The framework utilises Qwen3-VL for semantic task alignment and employs a latent consistency loss to bridge the gap between mimicry and world dynamics. This research, conducted as part of the MALESTRO project at the University of Malta, seeks to provide a scalable foundation for robust industrial automation and human-centric digital instruction.

Index Terms—Imitation Learning, World Models, Action Chunking Transformers (ACT), Visuomotor Policies, Industrial Digital Twins, Predictive Grounding, Embodied AI.

I. INTRODUCTION

The progressive loss of skilled labour is a global challenge whose impact is heavily felt across the industrial sector [1]. While the skills gap grows, research indicates that conventional onboarding methods often struggle to achieve durable learning outcomes. Findings on the forgetting curve indicate that a substantial proportion of newly learned information can be lost within the first 24 hours after instruction [2]. In contrast, immersive approaches such as Augmented Reality (AR) demonstrate an 82% reduction in operational errors compared to static manuals while increasing worker confidence by up to 275% [3], [4].

Despite these clear advantages, the creation of high-quality, action-labelled digital content remains a significant bottleneck in imitation learning. Unlike fields such as natural language processing (NLP) or 2D computer vision, the formulation of a standardised, Internet-scale corpus with explicit state-action pairs is not a natural passive process [5]. Whilst NLP and computer vision benefit from the fact that vast amounts of observational data (text, images, and video with basic meta-data) are already digitally available online, imitation learning requires bespoke, expert demonstrations with dense, precisely synchronised action labels which are expensive and difficult to scale [6], [7].

Typically, current imitation learning architectures focus primarily on solving the control problem. While these visuomotor policies excel at capturing high-frequency nuances in human

movement, they often operate as a “black-box” mapping from raw pixels to motor commands [8], [9]. Because these models lack an internal representation of environment dynamics, they are often unable to capture the real-world physics in which such motion is controlled. Consequently, the resulting policies are frequently brittle, replicating motions that have not undergone a physical constraint analysis, leading to failure when faced with minor environmental changes.

In response to this limitation, the latest literature has seen a strategic emergence of *Foundation World Models* (WFM) designed to learn the underlying physics of reality from vast video corpora. Rather than merely mimicking actions, these models learn to “simulate” the future, providing a predictive understanding of world dynamics [10]. This paper proposes a novel bridged framework that integrates a state-of-the-art (SOTA) visuomotor policy with an emerging world model to enable the generation of physically grounded action instructions. Established high-fidelity egocentric datasets are leveraged, alongside a multi-modal assessment strategy, to validate the proposed approach.

II. RESEARCH QUESTION AND AIM

The core challenge of this research lies in the architectural integration of predictive world models with high-precision action controllers. While WFMs provide a latent understanding of environment transitions, it remains unclear how this “internal simulation” can be effectively leveraged to regularise the output of a visuomotor policy. Without a formal method of alignment, the policy remains a passive mimic, unaware of the physical feasibility of the action sequences it generates. This work seeks to answer the following primary research question:

To what extent does the integration of a latent world model, trained on high-fidelity egocentric data, improve the robustness and physical grounding of an imitation learning policy ?

To address this, the central Aim of this research is to develop and validate a *World-Model Augmented ACT (W-ACT)* framework that utilises predictive grounding to enhance imitation learning. To achieve this, Table I outlines the high-level objectives of this research whilst also making reference the corresponding metric rows in Table II, providing explicit traceability from objectives to evaluation.

TABLE I
RESEARCH OBJECTIVES WITH EXPLICIT TRACEABILITY TO EVALUATION METRICS

ID	Research Objective	Eval Link
1	<i>Predictive World Modelling</i> : Learn task dynamics from egocentric observation in order to forecast future interaction states.	Metric 1
2	<i>Visuomotor Control Accuracy</i> : Learn a policy that reproduces expert action sequences with high spatial and temporal precision.	Metric 2
3	<i>Robust Task Execution</i> : Maintain task completion under perturbations and distribution shifts that reflect non ideal conditions.	Metric 3
4	<i>Deployment Efficiency</i> : Ensure inference latency remains suitable for interactive real-time operation.	Metric 4

III. BACKGROUND AND RELATED WORK

Semantic Alignment

A central debate in the literature regarding LfO is the architectural choice for semantic alignment, primarily contrasting monolithic *Vision-Language-Action* (VLA) models with decoupled *Vision-Language Models* (VLM). VLAs are monolithic transformers that tokenise vision, language, and actions into a single unified sequence. A representative SOTA example is *RT-2* [11], which encodes visual observations and actions as textual tokens, leveraging internet-scale web data to fine-tune the model. At inference time, predicted action tokens are de-tokenised into executable commands, enabling a closed-loop control process. However, while VLAs benefit from immense generalisation capabilities, they often lack the deterministic precision and high-frequency control required for complex manipulation.

In contrast, decoupled architectures utilise a VLM as a high-level semantic translator, delegating low-level execution to a specialist visuomotor policy. VLMs themselves are not end-to-end agents; they function as passive sequence generators rather than active decision-making agents [12]. This separation allows for the independent optimisation of task reasoning and physical execution, providing a pathway to integrate specialised controllers (see Section III). Despite these benefits, the VLM’s role as a “truth generator” is limited by susceptibility to semantic hallucinations and a lack of inherent confidence thresholds for ambiguous states. This necessitates a verification layer or confidence-weighted filtering to ensure that task embeddings (g) do not introduce a single point of failure through flawed high-level alignment.

Visuomotor Policies

The visuomotor policy serves as the controller, mapping high-dimensional observations to low-level motor commands. While modern Behaviour Cloning (BC) utilises temporal context to ensure trajectory robustness [13], a performance tension exists between two primary architectures. *Diffusion Policies*

treat action generation as an iterative denoising process, excelling at modelling complex multimodal distributions [14]. However, their iterative sampling can incur high inference latency (often tens to hundreds of milliseconds, depending on the number of denoising steps and hardware), which can be prohibitive for sub-20ms real-time industrial Augmented Reality (AR) requirements.

In contrast, *Action Chunking Transformers* (ACT) formulate imitation as a sequence modelling task conditioned on a Conditional Variational Autoencoder (CVAE) [15]. By predicting fixed-horizon “action chunks” ($a_{t:t+k}$), ACT reduces long-horizon compounding errors and enables high-frequency execution. In practice, ACT-style policies may additionally employ temporal smoothing or ensembling across overlapping predictions to improve trajectory continuity. Yet, while such smoothing can provide temporal stability, it does not in itself provide predictive grounding. Without an internal understanding of how the environment responds to these actions, the output remains a statistical extrapolation of demonstrations rather than a physically validated forecast. This creates a natural synergy for *Foundation World Models* (see Section III), which provide the latent physical reasoning necessary to regularise these action sequences.

Foundation World Models

World Foundation Models (WFMs) function as learned “simulators” that capture the underlying physics of an environment. They are pre-trained using self-supervised learning (SSL) on vast corpora of “actionless” internet video to bootstrap an understanding of physical laws (e.g., object permanence). The models are subsequently “action-conditioned” using smaller datasets to learn specific agent behaviours. The key innovation is that learning occurs in a latent space, focusing on “what happens next” rather than “what should I do?” [16].

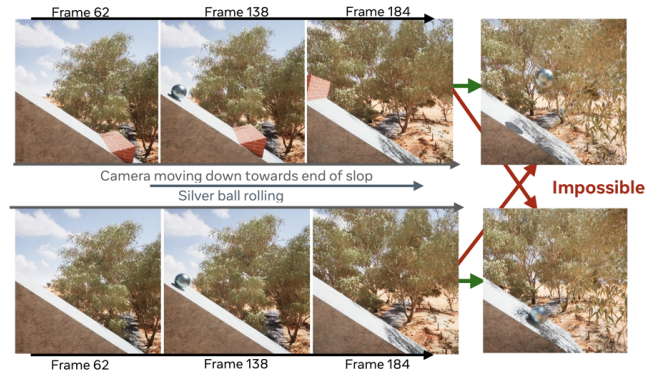


Fig. 1. A WFMs understanding of intuitive physics determination of possible or impossible future outcomes [17]

A core technical distinction in the literature is the modelling medium. Pixel-space generative world models (e.g., NVIDIA Cosmos) can provide high-fidelity visual roll-outs but are computationally intensive and prone to “visual noise” from task-irrelevant details [18]. In contrast, latent-space models,

such as *V-JEPA 2* [19], predict future latent embeddings for spatiotemporal patches (often conceptualised as “tubelets”), bypassing expensive pixel generation. This approach can reduce inference cost and emphasise physically salient dynamics.

Consequently, WFM demonstrate proficiency in dynamic navigation. As advocated by Yann LeCun, Joint-Embedding Predictive Architectures (JEPA) are a primary path towards “Machine Common Sense” by modelling hidden variables of world dynamics [20]. For this work, the WFM is used as a latent simulator to provide high-level spatiotemporal predictability and scene understanding, which aligns with the goal of providing predictive grounding for the ACT controller.

IV. PROPOSED IDEA

This work proposes a novel framework, *W-ACT* (*World-Model Augmented Action Chunking Transformer*), designed to bridge the gap between high-precision imitation control and physically grounded latent world modelling. The architecture is a decoupled system, leveraging SOTA models for specific tasks. To mitigate the labour- and time-intensive burden of bespoke data generation, an already established high-fidelity dataset is leveraged. The proposed dataset is *EgoDex* [21], which is open-source, extensive (hundreds of hours), and rich in egocentric video with corresponding 3D hand-pose annotations.

For semantic alignment and task embedding, an existing GPU cluster is utilised to launch a *Sovereign AI* model. The selected model is *Qwen3-VL* [22], a high-performing open-source VLM that offers flexibility and data retention benefits compared with external API-based services. *Qwen3-VL* extracts high-level task embeddings (g), which serve a dual purpose. It conditions the visuomotor policy on the intended goal and provides semantically aligned samples to guide the world model’s understanding of task-specific transitions.

The overall design of the *W-ACT* framework is detailed in Figure 2. The primary models utilised are the *V-JEPA 2* for world modelling and the *ACT* for visuomotor policy execution. The framework is initialised using the pre-trained encoder of *V-JEPA 2*, which already possesses a foundational “intuitive physics” prior bootstrapped from internet-scale video [19]. To adapt this to specific industrial workflows, the original encoder weights are frozen to preserve broad physical priors, whilst a predictive head is fine-tuned to create the *V-JEPA 2-AC* (*Action-Conditioned*) variant. This conditioned model learns to simulate how specific 3D hand poses and manipulations (a_t) alter the environment’s latent state (z_{t+1}).

Simultaneously, the *ACT Visuomotor Policy* processes current visual observations (O_t) and task embeddings to generate sequence-based “action chunks” ($\hat{a}_{t:t+k}$). Rather than executing these in an open loop, the proposed architecture feeds the generated chunks into the *V-JEPA 2-AC* latent simulator. This generates a forecasted future state ($\hat{z}_{t:t+k}$), which is then compared against the ground-truth test set via a composite loss function. This feedback mechanism provides the “predictive grounding” necessary to tune the *ACT* policy, ensuring that the

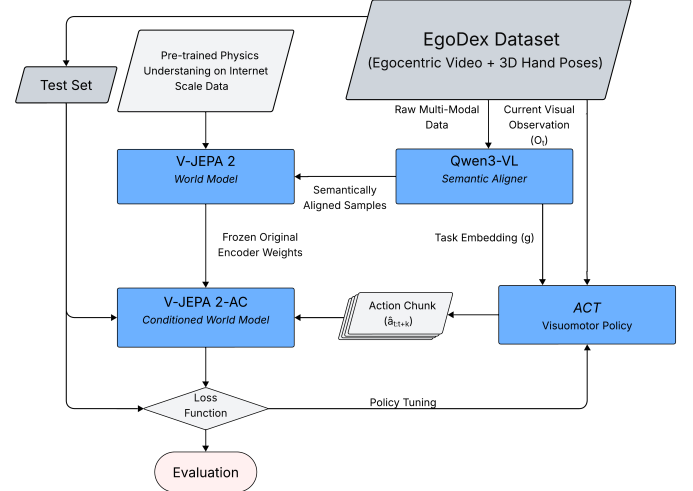


Fig. 2. The proposed *W-ACT* framework architecture, illustrating the flow from data ingestion to policy tuning via the world model feedback loop.

resulting motor commands are not mere statistical mimics of the expert data, but are validated against the predicted physical evolution of the scene.

V. EVALUATION AND BENCHMARKING

To evaluate the efficacy of the *W-ACT* framework, a multi-modal assessment strategy is employed that prioritises physical feasibility over mere imitation. A composite objective function is utilised for policy regularisation:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{ACT} + \lambda_2 \mathcal{L}_{KLD} + \lambda_3 \mathcal{L}_{ground} \quad (1)$$

where $\mathcal{L}_{ACT}$ represents the L1 reconstruction loss between the predicted action chunks and expert demonstrations. $\mathcal{L}_{KLD}$ denotes the Kullback-Leibler divergence, a type of statistical distance, used to regularise the CVAE latent-space against a prior distribution [15], [23]. The novel term, $\mathcal{L}_{ground}$, quantifies the *Latent Consistency Loss*. Intuitively, the world model is used to “simulate” how the scene representation should evolve under the policy’s predicted actions. The resulting predicted future latent state is then compared against the latent state obtained by encoding the corresponding future observation from *EgoDex*. The loss is implemented as a mean-squared error between predicted and target latent embeddings over a short prediction horizon [24].

In alignment with emerging world-model benchmarks such as WorldGym [25], the internal world model is validated through two distinct protocols: *Agreement on Validation Splits*, ensuring the latent simulator accurately reflects environmental transitions, and *End-Effector Control Sweeps*, testing the model’s predictive response to perturbed action inputs. To address the limitations of L1 trajectory matching, *Semantic Success Detection* is implemented using *Qwen3-VL* as a zero-shot judge [26]. This allows for the evaluation of functional

task success even in cases of minor trajectory deviations. W-ACT is benchmarked against a vanilla ACT baseline and a diffusion-based policy, using the criteria summarised in Table II.

TABLE II
PLANNED EVALUATION METRICS AND BENCHMARKING CRITERIA

ID	Category	Metric	Evaluation Intent
1	Predictive Fidelity	Latent Agreement (MSE)	Quantifies how well predicted latent futures match encoded latent states, as a proxy for learned interaction dynamics.
2	Control Precision	Trajectory Error (L1)	Measures deviation between generated trajectories and expert demonstrations to assess imitation accuracy.
3	Robustness	Success Rate (SR)	Estimates task completion under perturbations, with a VLM as a Judge for outcome validation when ground truth labels are limited.
4	Efficiency	Latency (ms)	Confirms suitability for real-time operation under an inference budget consistent with interactive industrial use.

VI. CONCLUSION

This research proposes the W-ACT framework as a strategic response to the scalability and robustness limitations currently hindering imitation learning in industrial applications. By bridging the gap between high-frequency visuomotor control and latent world modelling, the proposed approach is intended to demonstrate that "intuitive physics" can be effectively leveraged to regularise robotic policies, reducing their inherent brittleness. Through the integration of the V-JEPA 2 latent simulator and semantic evaluation via Qwen3-VL, this work provides a pathway for more resilient and physically grounded action generation.

The successful validation of this bridged architecture would mark a significant advancement in the deployment of digital twins for complex manipulation, offering a scalable foundation for both autonomous systems and human-centric industrial instruction. This research forms part of the holistic M.AI.ESTRO project at the University of Malta, conducted under the supervision of project leader Prof. Alexiei Dingli. Following such a framework, the respective predictions can be interfaced with live inference data to form an applicable action-guidance stream. This enables the real-time comparison of live egocentric observations against the world model's physical forecasts to generate immediate corrective overlays or robotic interventions.

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